

STATEMENT OF WORK – contractor Single Integrated Operational Picture, (SIOP) Improvement

1.0 Introduction/Background: As part of the Uncrewed Aerial Systems (UAS) Traffic Management Beyond Visual Line of Sight (UTM BVLOS) Sub-project under Air Traffic Management Exploration (ATM-X), NASA Langley is preparing to go through the FAA's newly developed Near-Term Approval Process (NTAP) as a BVLOS UAS operator, known as the Champion Operator (CO). As part of the NTAP process, the CO will partner with a supplemental data service provider to mitigate hazards associated with operating BVLOS. Langley is partnering with a surveillance Supplemental Data Service Provider (SDSP) to help utilize the NTAP process to help bring BVLOS services through and provide feedback on the process to the FAA. NASA will provide raw sensor feeds for an LSTAR radar and a uAvionix PingStation to be ingested into SIOP.

2.0 Scope of Work: Contractor provides a surveillance service, known as Single Integrated Operational Picture, or SIOP. SIOP ingests data feeds and provides a single integrated airspace display for multiple sensors, including radars, ADS-B sensors, and other surveillance devices. To go through the NTAP process, NASA would like this system to meet the ASTM F38 Standard Specification for Surveillance SDSP. The scope of this work would be an evaluation of the current SIOP system and the ASTM Surveillance standard to identify the gaps between the two, perform software updates required to meet these requirements and calculate the system measures of performances specified in the standard.

3.0 Objectives: The output of this work will be used as documentation for the NTAP submission with NASA Langley as the CO and contractor as the SDSP. This is an effort that the UTM BVLOS sub-project is pursuing to use the NTAP process to provide feedback to the FAA on it as well as acquire a Letter of Acceptance for BVLOS services. It will also allow for NASA Langley to conduct BVLOS operations outside of the current flight range.

4.0 Description of the Work/Contractor Tasks:

4.1 Task 1 – Current System Evaluation:

Contractor shall review the SIOP system and the requirements of ASTM F3623-23 “Standard for Specification for Surveillance SDSP” to identify which requirements SIOP currently meets, and which requirements will require more development. The contractor will also document how these requirements will be met through development of the SIOP system in support of Task 2. Requirements may require more support from the NASA to fully meet, such as information on the sensors being provided. These requirements should be clearly identified so that NASA can work with the contractor to provide the proper data to allow the requirement to be met.

4.2 Task 2 – System Update:

Contractor shall update the SIOP system to meet the ASTM F3623-23 “Standard for Specification for Surveillance SDSP” standard. This includes all development of required components for the system and calculation of measures of performance for SIOP to perform as a surveillance SDSP. This may involve using sensor information or procedural information from NASA, which will be provided based on the requirements identified in Task 4.1.

Once started, the contractor shall give weekly reports to NASA on the system updates which can be captured in a tele-meeting, email, or documented report.

4.3 Task 3 – System Performance Metrics Calculated:

Contractor shall calculate the metrics of the system that align with the ASTM Surveillance SDSP requires. These include:

- Mean time to recovery.
- Automatic recovery time.
- Mean time between failures with and without automatic recovery.
- Track update rate.
- Heartbeat rate.
- Probability of update.
- Probability of long gaps.
- Track accuracy, if changed from original sensors.
- Latency.
- Maximum tracks.

5.0 Deliverables:

Milestone	Date	Details
Contract award	--	Contract awarded and work begins
Document of ASTM standard compared to SIOP.	May 1, 2025	Document should include requirements currently met, requirements that need work to SIOP, requirements that need information from NASA, and how each requirement will be met.
Updates Started	May 1, 2025	Improvements on the SIOP system to meet the ASTM Surveillance SDSP standard begins.
Updates on progress	Weekly	Update on the progress of development, can be over telecon, email, or report.
Updates Complete	July 30, 2025	System is updated to a state where required components to meet the ASTM Surveillance SDSP standard are met and performance metrics can be calculated.
SIOP System Measures of Performance Documented.	August 15, 2025	Documentation of the measures of performance for Task 3 to map to the ASTM Surveillance SDSP standard.

Other Considerations:

Applicable Documents/Background:

Applicable Standards:

- ASTM F3623-23 – Standard Specification for Surveillance SDSP.
- ASTM F3442/ F3442M - Standard Specification for Detect and Avoid System Performance Requirements.
- FAA NTAP Service Evaluation Matrix.

Milestones: See deliverables above.

Government-furnished property - None

Government-furnished information – NASA to provide test data from surveillance sensors intended to be displayed on SIOP for testing the upgrades.